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Crawler-type work machine

Abstract

The bulldozer has a right first oil channel, a right second oil channel, and a right third oil channel. The right first oil channel guides a portion of the lubricating oil discharged by a hydraulic pump to a right steering brake. The right second first oil channel guides a portion of the lubricating oil discharged by the hydraulic pump to a right planetary gear mechanism. The right third oil channel guides at least a portion of the lubricating oil that has passed through the right planetary gear mechanism to a right steering brake. At least a portion of the planetary gear mechanism is disposed further inside than the right steering brake in a radial direction perpendicular to the center axis of an input shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National stage application of International Application No. PCT/JP2021/022729, filed on Jun. 15, 2021. This U.S. National stage application claims priority under 35 U.S.C. § 119(a) to Japanese Patent Application No. 2020-129004, filed in Japan on Jul. 30, 2020, the entire contents of which are hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a crawler-type work machine.

BACKGROUND ART

(3) In a crawler-type work machine (e.g., a bulldozer), power from an engine is transmitted through a transmission to left and right drive wheels so that left and right crawler belts are driven by the left and right drive wheels. In this type of crawler-type work machine, turning to the left and right is executed by controlling the hydraulic pressure of left and right steering clutches and left and right steering brakes provided in correspondence to the left and right drive wheels.

(4) Japanese Patent Laid-open No. 2017-71357 proposes a method for supplying, to a steering brake, a portion of the lubricating oil supplied from a hydraulic pump to a steering clutch, in order to use the lubricating oil efficiently.

SUMMARY

(5) While it is assumed that a planetary gear mechanism may be provided between an input shaft and an output shaft in order to make the steering clutch more compact, Japanese Patent Laid-open No. 2017-71357 does not examine how to efficiently use the lubricating oil when providing a planetary gear mechanism.

(6) An object of the present disclosure is to provide a crawler-type work machine that is able to use lubricating oil efficiently.

(7) A crawler-type work machine according to one aspect of the present disclosure has a planetary gear mechanism, a steering clutch, a steering brake, a hydraulic pump, a first oil channel, a second oil channel, and a third oil channel. The planetary gear mechanism is disposed between an input shaft and an output shaft. The steering clutch is configured to switch between transmitting and

blocking the rotation from the input shaft to the output shaft by the planetary gear mechanism. The steering brake is configured to brake the rotation of the output shaft. The hydraulic pump is configured to discharge a lubricating oil. The first oil channel is configured to guide a portion of the lubricating oil discharged by the hydraulic pump to the steering brake. The second oil channel is configured to guide a portion of the lubricating oil discharged by the hydraulic pump to the planetary gear mechanism. The third oil channel is configured to guide at least a portion of the lubricating oil that has passed through the planetary gear mechanism to the steering brake. At least a portion of the planetary gear mechanism is disposed further inside than the steering brake in a radial direction perpendicular to a center axis of the input shaft.

(8) According to the crawler-type work machine as in the present disclosure, the lubricating oil can be used efficiently.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a bulldozer that is an example of the crawler-type work machine.

(2) FIG. 2 is a cross-sectional configuration diagram of a power transmission system of the bulldozer.

(3) FIG. 3 is an outline system configuration diagram of the power transmission system of the bulldozer.

(4) FIG. 4 is a cross-sectional view illustrating a configuration in the vicinity of a lubricating oil channel.

(5) FIG. 5 is a schematic view for explaining the flow of the lubricating oil.

DESCRIPTION OF EMBODIMENTS

(6) (Configuration of Bulldozer 1)

(7) FIG. 1 is a perspective view of a bulldozer 1 that is an example of the crawler-type work machine. FIG. 2 is a cross-sectional configuration diagram of the power transmission system of the bulldozer 1. FIG. 3 is an outline system configuration diagram of the power transmission system of the bulldozer 1.

(8) The bulldozer 1 is equipped with: left and right travel devices 4L, 4R that respectively have left and right sprockets 2L, 2R and left and right crawler belts 3L, 3R; a blade 5 provided at a vehicle front part; and a ripper device 6 provided at a vehicle rear part.

(9) The bulldozer 1 is able to do work such as pushing earth with the blade 5, and work such as crushing and excavating with the ripper device 6.

(10) The bulldozer 1 includes an engine 10, a power transmission device 20, left and right planetary gear mechanisms 30L, 30R, left and right steering clutches 40L, 40R, left and right steering brakes 50L, 50R, left and right output shafts 60L, 60R, left and right final drive gears 70L, 70R, and a lubricating oil supply unit 80.

(11) The power transmission device 20 transmits power from the engine 10 to the left and right planetary gear mechanisms 30L, 30R. The power transmission device 20 includes a power take-off device 21, a torque converter 22, a transmission 23, a pinion 24, a bevel gear 25, and an input shaft 26.

(12) The power transmission device 21 transmits power from the engine 10 to the torque converter 22 and the lubricating oil supply unit 80. The torque converter 22 transmits the power of the engine 10 transmitted by the power take-off device 21 to the transmission 23 by means of a fluid. The transmission 23 changes the speed of the rotation motion transmitted from the torque converter 22. The transmission 23 is able to switch between forward travel and reverse travel. The transmission 23 is coupled to the pinion 24. Power from the transmission 23 is transmitted through the pinion 24

and the bevel gear **25** to the input shaft **26**.

(13) The left and right planetary gear mechanisms **30L**, **30R** are disposed between the input shaft **26** and the left and right output shafts **60L**, **60R**. The left and right planetary gear mechanisms **30L**, **30R** respectively have left and right ring gears **31L**, **31R**, left and right planetary gears **32L**, **32R**, left and right sun gears **33L**, **33R**, and left and right carriers **34L**, **34R**.

(14) The left and right ring gears **31L**, **31R** are coupled to the input shaft **26**. The left and right planetary gears **32L**, **32R** are respectively disposed inside the left and right ring gears **31L**, **31R** in a radial direction perpendicular to the center axis of the input shaft **26**. The left and right planetary gears **32L**, **32R** are respectively disposed between the left and right ring gears **31L**, **31R** and the left and right sun gears **33L**, **33R**. The left and right planetary gears **32L**, **32R** respectively mesh with the left and right ring gears **31L**, **31R** and the left and right sun gears **33L**, **33R**. The left and right sun gears **33L**, **33R** are rotatably attached to the input shaft **26**. The left and right sun gears **33L**, **33R** are respectively disposed inside the left and right planetary gears **32L**, **32R** in the radial direction. The left and right sun gears **33L**, **33R** are respectively coupled to the left and right steering clutches **40L**, **40R**. The left and right carriers **34L**, **34R** are respectively coupled to the left and right planetary gears **32L**, **32R** and the left and right output shafts **60L**, **60R**.

(15) The left and right steering clutches **40L**, **40R** are wet multiplate clutches that can be respectively engaged or disengaged with the left and right sun gears **33L**, **33R**. The left and right steering clutches **40L**, **40R** switch between respectively transmitting and blocking the rotational force from the input shaft **26** to the left and right output shafts **60L**, **60R** by means of the respective left and right planetary gear mechanisms **30L**, **30R**.

(16) Specifically, when the left steering clutch **40L** is coupled and the left sun gear **33L** is directly coupled to a first clutch gear **91**, the first clutch gear **91** cannot rotate (see below) and the left sun gear **33L** also cannot rotate. As a result, the rotation of the input shaft **26** is transmitted through the left ring gear **31L**, the left planetary gear **32L**, and the left carrier **34L** to the left output shaft **60L**. When the left steering clutch **40L** is disengaged and the left sun gear **33L** enters a freely rotating state, the rotation from the input shaft **26** to the left output shaft **60L** is blocked. Similarly, the transmission or blocking of the rotation from the input shaft **26** to the right output shaft **60R** is switched in accordance with the engagement or disengagement of the right steering clutch **40R**.

(17) As illustrated in FIG. 3, the first clutch gear **91** is attached to the left steering clutch **40L** and the first clutch gear **91** is coupled to an idler gear **93** and a pinion gear **94**. In addition, a second clutch gear **92** is attached to the right steering clutch **40R** and the second clutch gear **92** is coupled to the idler gear **93** and the pinion gear **94** via a first transfer gear **95**, an auxiliary shaft **96**, and a second transfer gear **97**. A pinion shaft **94a** is coupled to the pinion gear **94** and is fixed by a fixing member **98** in an unrotatable manner. Therefore, the rotational force from the left and right steering clutches **40L**, **40R** to the left and right sun gears **33L**, **33R** is not applied.

(18) The fixing member **98** is detachably fixed to an unillustrated housing. When the fixing member **98** is removed from the housing, the fixing member **98** can be removed from the pinion gear **94**. As a result, the fixing of the pinion gear **94** by the fixing member **98** can be released.

(19) When the fixing of the pinion gear **94** by the fixing member **98** is released, a turning motor (not illustrated) may be coupled to the pinion shaft **94a**. In this case, when the left and right steering clutches **40L**, **40R** are engaged and the rotational force of the turning motor is transmitted to the left and right sun gears **33L**, **33R** through the left and right steering clutches **40L**, **40R**, the left and right sun gears **33L**, **33R** rotate in the opposite directions at the same rotation speed. Consequently, a rotation speed difference is generated in the left and right output shafts **60L**, **60R** and the bulldozer **1** is able to turn gently to the left or right.

(20) The left and right steering brakes **50L**, **50R** respectively brake the rotation of the left and right output shafts **60L**, **60R**. The left and right steering brakes **50L**, **50R** are wet multiplate clutches that can be respectively engaged or disengaged with the left and right output shafts **60L**, **60R**.

(21) When the left steering brake **50L** is engaged, braking is applied to the rotation of the left

output shaft **60L**. As a result, the rotation of the left final drive gear **70L** coupled to the left sprocket **2L** is reduced. When the right steering brake **50R** is engaged, braking is applied to the rotation of the right output shaft **60R**. As a result, the rotation of the right final drive gear **70R** coupled to the right sprocket **2R** is reduced.

(22) The lubricating oil supply unit **80** has a hydraulic pump **81**, a lubricating oil channel **82**, and a lubricating oil tank **83**.

(23) The hydraulic pump **81** is driven by power from the engine **10** transmitted from the power take-off device **21**. The hydraulic pump **81** discharges the lubricating oil from the lubricating oil tank **83** to the lubricating oil channel **82**. The hydraulic pump **81** is a fixed displacement pump in the present embodiment.

(24) The lubricating oil channel **82** includes a first oil supply channel **100**, a second oil supply channel **110**, left and right first oil channels **101L**, **101R**, left and right second oil channels **102L**, **102R**, left and right third oil channels **103L**, **103R**, left and right fourth oil channels **104L**, **104R**, left and right fifth oil channels **105L**, **105R**, and left and right oil discharge channels **106L**, **106R**.

(25) The first oil supply channel **100** is continuous with the hydraulic pump **81** to the second oil supply channel **110** and the left and right first oil channels **101L**, **101R**. The first oil supply channel **100** supplies the lubricating oil discharged from the hydraulic pump **81** to the second oil supply channel **110** and the left and right first oil channels **101L**, **101R**.

(26) The left and right first oil channels **101L**, **101R** guide a portion of the lubricating oil discharged by the hydraulic pump **81** to the left and right steering brakes **50L**, **50R**.

(27) The second oil supply channel **110** is continuous with the first oil supply channel **100** and the inside of the input shaft **26**. The second oil supply channel **110** guides a portion of the lubricating oil discharged by the hydraulic pump **81** to the left and right second oil channels **102L**, **102R** and the left and right fourth oil channels **104L**, **104R**.

(28) The left and right second oil channels **102L**, **102R** respectively guide a portion of the lubricating oil discharged by the hydraulic pump **81** to the left and right planetary gear mechanisms **30L**, **30R**. The left and right third oil channels **103L**, **103R** respectively guide at least a portion of the lubricating oil that has passed through the left and right planetary gear mechanisms **30L**, **30R** to the left and right steering brakes **50L**, **50R**. In the present embodiment, the left and right third oil channels **103L**, **103R** respectively guide all of the lubricating oil that has passed through the left and right planetary gear mechanisms **30L**, **30R** to the left and right steering brakes **50L**, **50R**.

(29) The left and right fourth oil channels **104L**, **104R** respectively guide a portion of the lubricating oil discharged by the hydraulic pump **81** to the left and right steering clutches **40L**, **40R**. The left and right fifth oil channels **105L**, **105R** respectively guide at least a portion of the lubricating oil that has passed through the left and right steering brakes **40L**, **40R** to the left and right steering brakes **50L**, **50R**. Specifically, the left and right fifth oil channels **105L**, **105R** guide all of the lubricating oil that has passed through the left and right steering brakes **40L**, **40R** to the left and right steering brakes **50L**, **50R** via the left and right planetary gear mechanisms **30L**, **30R** and the left and right third oil channels **103L**, **103R**.

(30) In this way, all of the lubricating oil guided to the left and right planetary gear mechanisms **30L**, **30R** and to the left and right steering clutches **40L**, **40R** is guided respectively to the left and right steering brakes **50L**, **50R** via the left and right third oil channels **103L**, **103R**, and the left and right fifth oil channels **105L**, **105R** in the present embodiment.

(31) The left and right oil discharge channels **106L**, **106R** guide the lubricating oil that has passed through the left and right steering brakes **50L**, **50R** to the lubricating oil tank **83**.

(32) (Configuration in the Vicinity of the Lubricating Oil Channel **82**)

(33) FIG. **4** is a cross-sectional view illustrating a configuration in the vicinity of the lubricating oil channel **82**. A cross-section perpendicular to the center axis of the input shaft **26** is depicted in FIG. **4**. The configuration in the vicinity of the lubricating oil channel **82** is basically right-left symmetrical and the configuration on the right side will be explained below.

(34) The right steering clutch **40R** has a rotating member **41**, a clutch housing **42**, a plurality of clutch disks **43**, a plurality of clutch plates **44**, and a clutch piston **45**.

(35) The rotating member **41** is supported by the input shaft **26** and rotates around the center axis of the input shaft **26**. The clutch housing **42** is disposed so as to allow relative rotation with respect to the rotating member **41**. The clutch housing **42** encircles the outside of the clutch disks **43** and the clutch plates **44**. The clutch disks **43** are supported by the rotating member **41**. The clutch plates **44** are supported by the clutch housing **42**. The clutch disks **43** and the clutch plates **44** are disposed alternately in the axial direction parallel to the center axis of the input shaft **26**. The clutch piston **45** brings the clutch discs **43** and the clutch plates **44** into pressure contact when hydraulic fluid is supplied. As a result, the right steering clutch **40R** enters an engaged state.

(36) The right steering brake **50R** has a rotating member **51**, a brake housing **52**, a plurality of fixing plates **53**, a plurality of brake disks **54**, and a brake piston **55**.

(37) The rotating member **51** is fixed to the right output shaft **60R** and rotates around the center axis of the right output shaft **60R**. The brake housing **52** is able to rotate relatively with respect to the rotating member **51**. In the present embodiment, the brake housing **52** is fixed and the brake housing **52** itself does not rotate. The brake housing **52** encircles the outside of the fixing plates **53** and the brake disks **54**. The fixing plates **53** are attached to the brake housing **52**. The fixing plates **53** may move slightly in the axial direction with respect to the brake housing **52**. The brake disks **54** are supported by the rotating member **51**. The brake disks **54** may move slightly in the axial direction with respect to the rotating member **51**.

(38) The fixing plates **53** and the brake disks **54** are disposed alternately in the axial direction. The fixing plates **53** and the brake disks **54** are pressed together by an unillustrated spring (e.g., a disk spring or a coil spring). The brake piston **55** causes the fixing plates **53** and the brake disks **54** to separate from each other when hydraulic fluid is supplied. Consequently, the right steering brake **50R** enters a non-braking state.

(39) At least a portion of the right planetary gear mechanism **30R** is disposed further inside than the right steering brake **50R** in the radial direction. Specifically, the right ring gear **31R** is disposed further to the inside than the brake disks **54** in the radial direction. Moreover, the right planetary gear **32R** and the right sun gear **33R** disposed on the inside of the right ring gear **31R** are also disposed further to the inside than the brake disks **54** in the radial direction. Furthermore, a portion of the right carrier **34R** coupled to the right planetary gear **32R** and the right output shaft **60R** is also disposed further to the inside than the brake disks **54** in the radial direction.

(40) At least a portion of the right steering clutch **40R** is disposed further inside than the right steering brake **50R** in the radial direction. Specifically, the clutch disks **43** and the clutch plates **44** are disposed further to the inside than the brake disks **54** in the radial direction.

(41) The phrase “at least a portion of the member X is disposed further to the inside than the member Y in the radial direction” in the present description signifies that the minimum distance between the member X and the center axis is less than the minimum distance between the member Y and the center axis when viewing from the axial direction parallel to the center axis of the input shaft **26**. In this case, the position of the member X in the axial direction and the position of the member Y in the axial direction may partially or completely overlap each other or may not overlap each other.

(42) In the present embodiment, while the right steering brake **50R**, the right planetary gear mechanism **30R**, and the right steering clutch **40R** are disposed in said order from the right side, the disposition order in the axial direction may be changed as appropriate.

(43) Next, the flow of the lubricating oil will be explained. FIG. 5 is a schematic view for explaining the flow of the lubricating oil.

(44) The lubricating oil discharged from the hydraulic pump **81** is distributed to the right first oil channel **101R**, the right second oil channel **102R**, and the right fourth oil channel **104R** via the first oil supply channel **100** and the second oil supply channel **110**. The distribution ratio to the oil

channels may be adjusted as appropriate by changing the opening radius of each oil channel.

(45) The lubricating oil distributed to the right fourth oil channel **104R** is guided from the right fifth oil channel **105R** to the right planetary gear mechanism **30R** by centrifugal force after the right steering clutch **40R** has been lubricated.

(46) After the right planetary gear mechanism **30R** has been lubricated, the lubricating oil distributed to the right second oil channel **102R** is guided from the right third oil channel **103R** to the right steering brake **50R** by centrifugal force along with the lubricating oil guided from the right fifth oil channel **105R** to the right planetary gear mechanism **30R**.

(47) After the right steering brake **50R** has been lubricated with the lubricating oil guided from the right third oil channel **103R** to the right steering brake **50R**, the lubricating oil distributed to the right first oil channel **101R** is guided from the right oil discharge passage **106R** to the lubricating oil tank **83** by centrifugal force.

(48) The bulldozer **1** includes a right first oil channel **101R**, a right second oil channel **102R**, and a right third oil channel **103R**. The right first oil channel **101R** guides a portion of the lubricating oil discharged by the hydraulic pump **81** to the right steering brake **50R**. The right second oil channel **102R** guides a portion of the lubricating oil discharged by the hydraulic pump **81** to the right planetary gear mechanism **30R**. The third oil channel **103R** guides at least a portion of the lubricating oil that has passed through the right planetary gear mechanism **30R** to the right steering brake **50R**. At least a portion of the planetary gear mechanism **30R** is disposed further inside than the right steering brake **50R** in the radial direction perpendicular to the center axis of the input shaft **26**.

(49) Therefore, because the lubricating oil can be guided from the right planetary gear mechanism **30R** to the right steering brake **50R** with centrifugal force via the third oil channel **103R**, the lubricating oil used for lubricating the right planetary gear mechanism **30R** can also be used for lubricating the right steering brake **50R**. As a result, the lubricating oil supplied from the hydraulic pump **81** can be used efficiently and the capacity of the hydraulic pump **81** can be reduced. This result can be obtained in the same way for the left steering brake **50L**.

(50) The bulldozer **1** includes the right fourth oil channel **104R** and the right fifth oil channel **105R**. The right fourth oil channel **104R** guides a portion of the lubricating oil discharged by the hydraulic pump **81** to the right steering clutch **40R**. The fifth oil channel **105R** guides at least a portion of the lubricating oil that has passed through the right steering clutch **40R** to the right steering brake **50R**. At least a portion of the right steering clutch **40R** is disposed further inside than the right steering brake **50R** in the radial direction.

(51) Therefore, because the lubricating oil can be guided from the right steering clutch **40R** to the right steering brake **50R** by centrifugal force via the fifth oil channel **105R**, the lubricating oil used for lubricating the right steering clutch **40R** can also be used for lubricating the right steering brake **50R**. As a result, the lubricating oil supplied from the hydraulic pump **81** can be used efficiently and the capacity of the hydraulic pump **81** can be reduced. This result can be obtained in the same way for the left steering brake **50L**.

(52) The present disclosure is not limited to the above embodiment and various changes and modifications may be made without departing from the spirit of the disclosure.

(53) In the present embodiment, although the left and right third oil channels **103L**, **103R** respectively guide all of the lubricating oil that has passed through the left and right planetary gear mechanisms **30L**, **30R** to the left and right steering brakes **50L**, **50R**, the present disclosure is not limited thereto. The left and right third oil channels **103L**, **103R** may also respectively guide only a portion of the lubricating oil that has passed through the left and right planetary gear mechanisms **30L**, **30R** to the left and right steering brakes **50L**, **50R**. In this case, the remaining portion of the lubricating oil that has passed through the left and right planetary gear mechanisms **30L**, **30R** may be returned directly to the lubricating oil tank **83**.

(54) In the present embodiment, although the left and right fifth oil channels **105L**, **105R**

respectively guide all of the lubricating oil that has passed through the left and right steering clutches **40L**, **40R** to the left and right steering brakes **50L**, **50R**, the present disclosure is not limited thereto. The left and right fifth oil channels **105L**, **105R** may also respectively guide only a portion of the lubricating oil that has passed through the left and right steering clutches **40L**, **40R** to the left and right steering brakes **50L**, **50R**. In this case, the remaining portion of the lubricating oil that has passed through the left and right steering clutches **40L**, **40R** may be returned directly to the lubricating oil tank **83**.

(55) In the present embodiment, although the left and right fifth oil channels **105L**, **105R** are respectively connected from the left and right steering clutches **40L**, **40R** to the left and right planetary gear mechanisms **30L**, **30R**, the present disclosure is not limited thereto. For example, the left and right fifth oil channels **105L**, **105R** may also be respectively connected directly from the left and right steering clutches **40L**, **40R** to the left and right steering brakes **50L**, **50R**.

Alternatively, the left and right fifth oil channels **105L**, **105R** may also be respectively connected directly from the left and right steering clutches **40L**, **40R** to the lubricating oil tank **83**.

Claims

1. A crawler-type work machine comprising: an engine; a transmission configured to transmit power from the engine; an input shaft arranged to receive power transmitted from the transmission; a planetary gear mechanism disposed between the input shaft and an output shaft; a steering clutch configured to switch between transmitting and blocking a rotational force from the input shaft to the output shaft by the planetary gear mechanism; a steering brake configured to brake a rotation of the output shaft; a hydraulic pump configured to discharge a lubricating oil; a first oil channel configured to guide a portion of the lubricating oil discharged by the hydraulic pump to the steering brake; a second oil channel configured to guide a portion of the lubricating oil discharged by the hydraulic pump to the planetary gear mechanism; a third oil channel configured to guide at least a portion of the lubricating oil that has passed through the planetary gear mechanism to the steering brake; a fourth oil channel configured to guide a portion of the lubricating oil discharged by the hydraulic pump to the steering clutch; and a fifth oil channel configured to guide at least a portion of the lubricating oil that has passed through the steering clutch to the steering brake, at least a portion of the planetary gear mechanism being disposed further inside than the steering brake in a radial direction perpendicular to a center axis of the input shaft.
 2. The crawler-type work machine according to claim 1, wherein the planetary gear mechanism has a ring gear coupled to the input shaft, a planetary gear disposed inside the ring gear in the radial direction, a sun gear disposed inside the planetary gear in the radial direction and coupled to the steering clutch, and a carrier coupled to the planetary gear and the output shaft; the steering brake has a rotating member configured to rotate with the output shaft, a brake housing configured to be able to rotate relatively with respect to the rotating member, a plurality of fixing plates attached to the brake housing, and a plurality of brake disks attached to the rotating member; and the ring gear is disposed further inside than the plurality of brake disks in the radial direction.
 3. The crawler-type work machine according to claim 1, wherein at least a portion of the steering clutch is disposed further inside than the steering brake in the radial direction.
 4. The crawler-type work machine according to claim 2, wherein at least a portion of the steering clutch is disposed further inside than the steering brake in the radial direction.
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